ISIN 325 - Competition

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ISIN 325 - Database Security

1) Display how many purchases and the total amount purchased by each customer using one SQL statement.

```sql
Select C.CustomerID, C.FName, C.LName, Count(P.PurchaseNbr) as PurchaseNbr, SUM(PD.qty) as TotalItems
From Customer C, Purchase P, PurchaseDetail PD
Where C.CustomerID = P.CustomerID
and P.PurchaseNbr = PD.PurchaseNbr
Group By C.CustomerID, C.FName, C.LName
```

95 % 1 0

Results Messages

	CustomerID	FName	LName	PurchaseNbr	TotalItems
1	1	Paul	Powell	5	11.00
2	2	Tyler	Young	2	2.00
3	3	Anna	Adams	4	5.00
4	4	Julie	Watson	2	3.00
5	5	Ella	Hughes	4	5.00
6	6	Emma	Ward	2	3.00
7	7	Dorothy	Jones	4	6.00
8	8	Emily	Bell	2	6.00
9	9	Carol	Scott	2	2.00
10	10	Katherine	Gray	4	7.00
11	11	Justin	Clark	2	4.00
12	12	Ryan	Alexander	2	3.00
13	13	Katherine	Ramirez	2	3.00
14	14	Frank	Miller	2	3.00
15	15	Jennifer	Bailey	2	6.00

2) Display how many customers are in each city using one SQL statement.

```sql
Select S.City, Count(*)CustomerID
From Customer C, Purchase P, Store S
Where C.CustomerID = P.CustomerID
and P.StoreID = S.StoreID
Group By S.City
```

95 % No issues found

Results Messages

	City	CustomerID
1	Grand Rapids	13
2	Grandville	4
3	Walker	3

3) Display all the sales each product is tied to, with the output sorted by product and category.

```sql
Select C.Name as Category, P.Name as Product, PD.PurchaseNbr, PD.Qty
from Category C, Product P, PurchaseDetail PD
Where P.CategoryID = C.CategoryID
and PD.ProductID = P.ProductID
Order by C.Name, P.Name
```

95 % No issues found

Results Messages

	Category	Product	PurchaseNbr	Qty
1	Bakery	Product 13	1	3.00
2	Bakery	Product 13	8	4.00
3	Bakery	Product 13	15	5.00
4	Bakery	Product 3	14	2.00
5	Bakery	Product 3	6	1.00
6	Bakery	Product 3	1	1.00
7	Breakfast Foods	Product 8	8	2.00
8	Breakfast Foods	Product 8	12	1.00
9	Breakfast Foods	Product 8	18	2.00
10	Canned Fruits	Product 10	13	2.00
11	Canned Fruits	Product 10	5	1.00
12	Canned Vegetables	Product 9	3	1.00
13	Canned Vegetables	Product 9	16	2.00
14	Canned Vegetables	Product 9	10	1.00
15	Dairy Products	Product 12	14	1.00
16	Dairy Products	Product 12	6	2.00
17	Dairy Products	Product 12	19	2.00
18	Dairy Products	Product 2	19	2.00
19	Dairy Products	Product 2	2	1.00
20	Dairy Products	Product 2	11	3.00
21	Fresh Produce	Product 1	17	1.00
22	Fresh Produce	Product 1	1	2.00
23	Fresh Produce	Product 1	9	1.00
24	Fresh Produce	Product 11	5	1.00
25	Fresh Produce	Product 11	18	1.00

4) Display all the products in each category.

```
Select P.Name as ProductName, C.Name as CategoryName
From Product P, Category C
Where P.CategoryID = C.CategoryID
```

95 % 2 0

Results | Messages

	ProductName	CategoryName
1	Product 1	Fresh Produce
2	Product 2	Dairy Products
3	Product 3	Bakery
4	Product 4	Meat
5	Product 5	Poultry
6	Product 6	Seafood
7	Product 7	Frozen Foods
8	Product 8	Breakfast Foods
9	Product 9	Canned Vegetables
10	Product 10	Canned Fruits
11	Product 11	Fresh Produce
12	Product 12	Dairy Products
13	Product 13	Bakery
14	Product 14	Meat
15	Product 15	Poultry
16	Product 16	Seafood
17	Product 17	Frozen Foods
18	Product 18	Breakfast Foods
19	Product 19	Canned Vegetables
20	Product 20	Canned Fruits
21	Product 21	Fresh Produce
22	Product 22	Dairy Products
23	Product 23	Bakery
24	Product 24	Meat
25	Product 25	Poultry
26	Product 26	Seafood
27	Product 27	Frozen Foods
28	Product 28	Breakfast Foods

5) Display the product name and category sorted by category, then by product.

```sql
Select P.Name as ProductName, C.Name as CategoryName
From Product P, Category C
Where P.CategoryID = C.CategoryID
Order by C.Name, P.Name;
```

95 % No issues found

Results Messages

	ProductName	CategoryName
1	Product 13	Bakery
2	Product 23	Bakery
3	Product 3	Bakery
4	Product 33	Bakery
5	Product 43	Bakery
6	Product 53	Bakery
7	Product 63	Bakery
8	Product 73	Bakery
9	Product 83	Bakery
10	Product 93	Bakery
11	Product 18	Breakfast Foods
12	Product 28	Breakfast Foods
13	Product 38	Breakfast Foods
14	Product 48	Breakfast Foods
15	Product 58	Breakfast Foods
16	Product 68	Breakfast Foods
17	Product 78	Breakfast Foods
18	Product 8	Breakfast Foods
19	Product 88	Breakfast Foods
20	Product 98	Breakfast Foods
21	Product 10	Canned Fruits
22	Product 100	Canned Fruits
23	Product 20	Canned Fruits
24	Product 30	Canned Fruits
25	Product 40	Canned Fruits
26	Product 50	Canned Fruits
27	Product 60	Canned Fruits
28	Product 70	Canned Fruits

6) Display a count of the number of customers in each city, sorted by city.

```sql
Select S.City, Count(*)CustomerID
From Customer C, Purchase P, Store S
Where C.CustomerID = P.CUstomerID
and P.StoreID = S.StoreID
Group by S.City
```

95 % 1 0

Results Messages

	City	CustomerID
1	Grand Rapids	13
2	Grandville	4
3	Walker	3

7) Insert a new category "Health and Beauty Aids" with an appropriate description into the category table. (show me the sql and result set).

```sql
Insert into Category
    (Name, Descr)
Values
    ('Health and Beauty Aids', 'health and beauty products from Meijer')
Go
Select * from Category
```

95 % No issues found

Results Messages

	CategoryID	Name	Descr
78	78	Board Games	Family and strategy board games
79	79	Video Games	Video game software and accessories
80	80	Electronics	Consumer electronic devices
81	81	TV & Home Theater	Televisions and theater equipment
82	82	Computers	Desktop and laptop computers
83	83	Computer Accesso...	Keyboards, mice, and peripherals
84	84	Mobile Phones	Smartphones and mobile devices
85	85	Phone Accessories	Cases, chargers, and phone accessories
86	86	Cameras	Digital cameras and accessories
87	87	Audio Equipment	Speakers, headphones, and audio gear
88	88	Automotive Supplies	Car care and maintenance products
89	89	Motor Oil	Engine oils and lubricants
90	90	Car Electronics	Vehicle electronics and accessories
91	91	Tools	Hand tools and tool sets
92	92	Power Tools	Electric and battery powered tools
93	93	Hardware	Nuts, bolts, screws, and fasteners
94	94	Paint Supplies	Paint, brushes, and painting tools
95	95	Lawn & Garden	Gardening tools and lawn care products
96	96	Outdoor Furniture	Patio and outdoor furniture
97	97	Grills & BBQ	Outdoor grills and barbecue accessories
98	98	Camping Gear	Camping equipment and supplies
99	99	Sports Equipment	Equipment for various sports
100	100	Fitness Equipment	Home exercise and workout equipment
101	101	Bicycles	Bikes and cycling accessories
102	103	Health and Beauty ...	health and beauty products from Meijer

8) Insert records that indicate the purchases Judy Denth and Idris Elba made from Health and Beauty Aids and produce a report. Include the SQL to perform the inserts as well.

```sql
Insert into Customer
    (FName, LName)
Values
    ('Judy','Denth'),
    ('Idris','Elba')
Insert into Purchase
    (CustomerID, StoreID)
Values
    (501,1),
    (502,2)
Insert into PurchaseDetail
    (PurchaseNbr, ProductID, Qty)
Values
    (25,1,1),
    (26,2,5)
Select Cust.FName, Cust.LName, C.Name as Category, P.Name as Product, PD.Qty
From Customer Cust, Purchase Pur, PurchaseDetail PD, Product P, Category C
Where Cust.CustomerID = Pur.CustomerID
and Pur.PurchaseNbr = PD.PurchaseNbr
and PD.ProductID = P.ProductID
and P.CategoryID = C.CategoryID
and C.CategoryID = 103
and LName IN ('Denth', 'Elba')
Order by Cust.LName
```

95 % No issues found

Results Messages

	FName	LName	Category	Product	Qty
1	Judy	Denth	Health and Beauty Aids	Product 1	1.00
2	Idris	Elba	Health and Beauty Aids	Product 2	5.00

9) Display the category, purchase date and amount for each recipient include LName, FName) for each category sorted by Lname that have purchased at least 4 products.

```sql
Select C.Name as Category, Pur.DateTime as PurchaseDate, Sum(P.Price * PD.Qty) as Amount, Cust.LName, Cust.FName
From Customer Cust, Purchase Pur, PurchaseDetail PD, Product P, Category C
Where Cust.CustomerID = Pur.CustomerID
and Pur.PurchaseNbr = PD.PurchaseNbr
and PD.ProductID = P.ProductID
and P.CategoryID = C.CategoryID
Group by C.Name, Pur.DateTime, Cust.LName, Cust.FName
Order by Cust.LName
```

95 % No issues found Ln: 8, Ch: 20 TABS CRLF Windows 1

Results Messages

	Category	PurchaseDate	Amount	LName	FName
1	Canned Vegetables	2025-11-05 14:08:00.000	14.000000	Adams	Anna
2	Frozen Foods	2025-11-05 14:08:00.000	24.000000	Adams	Anna
3	Health and Beauty Aids	2025-12-13 13:00:00.000	6.000000	Adams	Anna
4	Poultry	2025-12-13 13:00:00.000	20.000000	Adams	Anna
5	Breakfast Foods	2025-12-01 14:44:00.000	13.000000	Alexander	Ryan
6	Poultry	2025-12-01 14:44:00.000	20.000000	Alexander	Ryan
7	Bakery	2025-12-09 16:00:00.000	90.000000	Bailey	Jennifer
8	Meat	2025-12-09 16:00:00.000	19.000000	Bailey	Jennifer
9	Bakery	2025-11-05 15:00:00.000	72.000000	Bell	Emily
10	Breakfast Foods	2025-11-05 15:00:00.000	26.000000	Bell	Emily
11	Health and Beauty Aids	2025-12-01 10:10:00.000	21.000000	Clark	Justin
12	Meat	2025-12-01 10:10:00.000	9.000000	Clark	Justin
13	Health and Beauty Aids	2026-04-28 10:23:13.810	6.000000	Denth	Judy
14	Health and Beauty Aids	2026-04-28 10:23:13.810	35.000000	Elba	Idris

10) Display a count of the number of products for each category using one SQL statement.

```sql
Select C.Name as Category, Count(P.ProductID) as ProductCount
From Category C, Product P
Where C.CategoryID = P.CategoryID
Group by C.Name
```

95 % No issues found

Results Messages

	Category	ProductCount
1	Bakery	9
2	Breakfast Foods	10
3	Canned Fruits	10
4	Canned Vegetables	10
5	Dairy Products	9
6	Fresh Produce	9
7	Frozen Foods	10
8	Health and Beauty Aids	3
9	Meat	10
10	Poultry	10
11	Seafood	10

11) Display the active logins in SQL Server.

```
1  exec sp_who
```

95 % No issues found

Results | Messages

	spid	ecid	status	loginame	hostname	blk	dbname	cmd	request_id
32	34	0	sleeping	sa		0	master	TASK MANAGER	0
33	37	0	sleeping	sa		0	master	TASK MANAGER	0
34	39	0	background	sa		0	master	TRACE QUEUE TASK	0
35	40	0	background	sa		0	NULL	RECEIVE	0
36	41	0	background	sa		0	master		0
37	42	0	background	sa		0	master	CHECKPOINT	0
38	43	0	background	sa		0	master	BRKR EVENT HNDLR	0
39	44	0	background	sa		0	master	BRKR TASK	0
40	45	0	background	sa		0	master	BRKR TASK	0
41	46	0	background	sa		0	master	BRKR TASK	0
42	49	0	sleeping	sa		0	master	TASK MANAGER	0
43	50	0	sleeping	sa		0	master	TASK MANAGER	0
44	51	0	sleeping	NT SE...	2-IN-1	0	master	AWAITING COMMAND	0
45	52	0	sleeping	sa		0	master	TASK MANAGER	0
46	53	0	background	sa		0	NULL	PARALLEL REDO TASK	0
47	54	0	background	sa		0	NULL	PARALLEL REDO TASK	0
48	55	0	runnable	2-IN-1\...	2-IN-1	0	Meijer	SELECT	0
49	56	0	sleeping	sa		0	master	TASK MANAGER	0
50	57	0	sleeping	sa		0	master	TASK MANAGER	0
51	58	0	sleeping	sa		0	master	TASK MANAGER	0
52	59	0	sleeping	sa		0	master	TASK MANAGER	0
53	60	0	sleeping	sa		0	master	TASK MANAGER	0
54	61	0	sleeping	sa		0	master	TASK MANAGER	0
55	62	0	sleeping	sa		0	master	TASK MANAGER	0
56	63	0	sleeping	sa		0	master	TASK MANAGER	0
57	66	0	sleeping	sa		0	master	TASK MANAGER	0
58	69	0	sleeping	2-IN-1\...	2-IN-1	0	master	AWAITING COMMAND	0
59	72	0	sleeping	2-IN-1\...	2-IN-1	0	Meijer	AWAITING COMMAND	0
60	74	0	runnable	2-IN-1\...	2-IN-1	0	Meijer	SELECT	0
61	77	0	sleeping	sa		0	master	TASK MANAGER	0

12) Demonstrate how to terminate a connection to SQL Server.

```
kill 69
```

```
%    1  0
Messages
Commands completed successfully.

Completion time: 2026-04-25T14:33:42.6038534-04:00
```

14) Create a trigger that performs a cascading delete in a second table when a record is deleted from a table.

```
Create Trigger PurchasePD
On Purchase
After Delete
As
Begin
    Delete from PurchaseDetail
    Where PurchaseNbr in (Select PurchaseNbr from deleted);
End
```

```
%    No issues found
Messages
Commands completed successfully.

Completion time: 2026-04-30T09:54:16.2287746-04:00
```

15) SQL to backup a database.

```sql
Backup Database Meijer
To Disk = 'C:\Program Files\Microsoft SQL Server\MSSQL17.SQLEXPRESS\MSSQL\Backup\Meijer.bak'
With Init, Name = 'Meijer';
```

No issues found Ln: 2, Ch: 81 TA

essages

```
Processed 1128 pages for database 'Meijer', file 'Meijer' on file 1.
Processed 1 pages for database 'Meijer', file 'Meijer_log' on file 1.
BACKUP DATABASE successfully processed 1129 pages in 0.080 seconds (110.174 MB/sec).

Completion time: 2026-04-29T20:51:42.9614205-04:00
```

16) SQL to restore a database.

```sql
Restore Database Meijer
From disk = 'C:\Program Files\Microsoft SQL Server\MSSQL17.SQLEXPRESS\MSSQL\Backup\Meijer.bak'
With recovery
```

No issues found Ln: 1, Ch: 1 TA

ssages

```
Processed 1128 pages for database 'Meijer', file 'Meijer' on file 1.
Processed 1 pages for database 'Meijer', file 'Meijer_log' on file 1.
RESTORE DATABASE successfully processed 1129 pages in 0.034 seconds (259.234 MB/sec).

Completion time: 2026-04-29T20:54:48.9016411-04:00
```

17) SQL to create a user and assign to a role in a database.

```sql
Create Login WyattKarnes With Password = 'Password123'

Go
Use Meijer;
Create User WyattKarnes For Login WyattKarnes;

Go
Alter Role db_datareader Add Member WyattKarnes;
```

% No issues found

Messages

```
Commands completed successfully.

Completion time: 2026-04-29T20:40:25.5489784-04:00
```